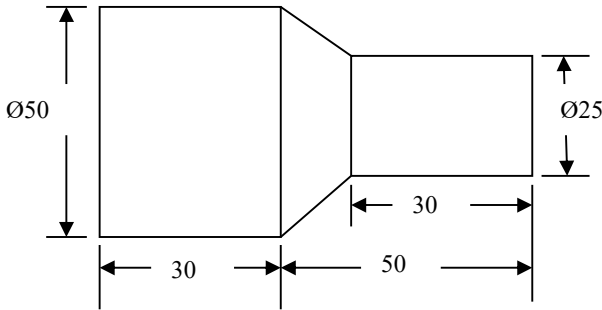
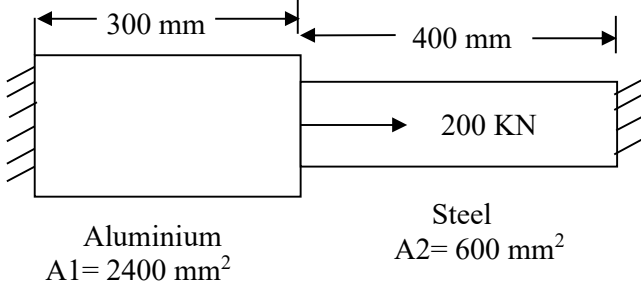


DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE Supplementary Winter Examination – 2023 Course: B. Tech. Branch : Mechanical Engineering & Allied Semester :VII Subject Code & Name: BTMEC702 CAD/CAM Max Marks: 60 Date:04-01-24 Duration: 3 Hr.			
Instructions to the Students: 1. All the questions are compulsory. 2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question. 3. Use of non-programmable scientific calculators is allowed. 4. Assume suitable data wherever necessary and mention it clearly.			
		(Level/CO)	Marks
Q. 1	Solve Any Two of the following.		12
A)	Briefly explain windowing, viewport and clipping in computer graphics.	CO1	6
B)	A triangle ABC with vertices A (20, 20), B (80, 20), and C (20, 80) is to be scaled by factor of 0.5 about a point X (40, 40). Determine composite transformation matrix and coordinates of vertices of scaled triangle.	CO2	6
C)	A triangle ABC with vertices A (2, 5), B (6, 7), and C (2, 7) is to be reflected about the line $Y = 0.5X + 3$. Determine composite transformation matrix and coordinates of vertices of scaled triangle.	CO2	6
Q.2	Solve Any Two of the following.		12
A)	Plot the Bezier Curve having endpoints $P_0 = (1,3)$ and $P_3(7,2)$. The other control points are $P_1 (5,6)$ and $P_2(6,0)$. Plot for values for $u=0(0.1)1$, if the characteristic polygon is drawn in sequence $P_0-P_1-P_2-P_3$.	CO4	6
B)	Explain with neat sketches surface modeling technique of geometric modeling.	CO2	6
C)	What is wireframe modeling? write its advantages and limitations	CO3	6

Q. 3	Solve Any Two of the following.		12
A)	Write a manual part program for a plain and taper turning the forged bar of 110 mm diameter as per the drawing shown. 	CO5	6
B)	Classify numerical control system based on: 1. Type of control systems 2. Type of motion control 3. Number of axes Briefly discuss advantages and disadvantages of any one NC system.	CO5	6
C)	List three basic steps of MRP. Discuss any one in detail.	CO6	6
Q.4	Solve Any Two of the following.		12
A)	List minimum six properties of global stiffness matrix.	CO6	6
B)	What do you mean by mesh generation? What are various mesh generation techniques?	CO6	6
C)	A stepped bimetallic bar of aluminium ($E = 70 \times 10^3 \text{ N/mm}^2$) and steel ($E = 200 \times 10^3 \text{ N/mm}^2$) is subjected to an axial load of 200 kN, as shown in fig, using finite element method determine: Nodal displacements. (All dimensions are in mm)  Aluminium $A_1 = 2400 \text{ mm}^2$ Steel $A_2 = 600 \text{ mm}^2$	CO6	6
Q. 5	Solve Any One of the following.		12
A)	Discuss types of flexible manufacturing system layouts.	CO7	6
B)	What is computer integrated manufacturing (CIM)? List its benefits.	CO8	6
C)	Briefly explain Retrieval & Generative CAPP systems.	CO9	6
	*** End ***		